

BK7259 BEKEN KWS Library User Manual

This document describes how to use the BK7259 KWS prebuilt libraries and the TensorFlow Lite model files in this directory.

Library Selection

The SDK selects the library automatically in `sdk/ap/components/bk_asr_service/CMakeLists.txt`.

- `CONFIG_BEKEN_KWS_MODEL_FROM_UDISK=n`: links `libkws_model.a`.
- `CONFIG_BEKEN_KWS_MODEL_FROM_UDISK=y`: links `libkws_model_vfs.a`.

Built-In Mode

Built-in mode uses `libkws_model.a`.

In this mode the KWS model arrays are compiled into the static library. No external `.tflite` file is needed at runtime.

Recommended use:

```
CONFIG_ASR_SERVICE=y
CONFIG_TFLITE_MICRO=y
CONFIG_BEKEN_KWS=y
CONFIG_BEKEN_KWS_MODEL_FROM_UDISK=n
CONFIG_NPU=y
CONFIG_NPU_CACHE=y
```

VFS Mode

VFS mode uses `libkws_model_vfs.a`.

In this mode the built-in model arrays are disabled to save flash. The runtime loads the `.tflite` model files from VFS by calling `bk_tflite_asr_set_model_from_file()`.

Recommended use:

```
CONFIG_ASR_SERVICE=y
CONFIG_TFLITE_MICRO=y
CONFIG_BEKEN_KWS=y
CONFIG_BEKEN_KWS_MODEL_FROM_UDISK=y
CONFIG_BEKEN_KWS_MODEL_BUFFER_USE_PSRAM=y
CONFIG_NPU=y
CONFIG_NPU_CACHE=y
CONFIG_VFS=y
CONFIG_FATFS=y
```

If the board uses SD-NAND or SD card, also enable the board-specific SDIO/FATFS configuration required by the project.

VFS Model File Names

The runtime file names are fixed by `audio_engine.c`:

```
1:/kws_model/bk_kws_wakeup.tflite
1:/kws_model/bk_kws_commands.tflite
```

When preparing a U-disk, SD card, or SD-NAND partition, create this directory:

```
1:/kws_model/
```

Then copy the two model files to it:

```
bk7259/bk_kws_wakeup.tflite -> 1:/kws_model/bk_kws_wakeup.tflite
bk7259/bk_kws_commands.tflite -> 1:/kws_model/bk_kws_commands.tflite
```

Do not change these runtime file names unless the application code path in `audio_engine.c` is changed at the same time.

Common Issues

Build Links The Wrong Library

Check `CONFIG_BEKEN_KWS_MODEL_FROM_UDISK`.

- `n` links `libkws_model.a`.
- `y` links `libkws_model_vfs.a`.

VFS Model Open Failed

Check the storage mount path and exact file names. The code expects:

```
1:/kws_model/bk_kws_wakeup.tflite
1:/kws_model/bk_kws_commands.tflite
```

NPU Or KWS Init Failed

Check:

- `CONFIG_NPU=y`
- `CONFIG_NPU_CACHE=y`
- arena and scratch buffers are 32-byte aligned
- PSRAM/HSRAM heap has enough free memory